Xinyun (Mark) Zhou

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Education

University of Wisconsin–Madison — Madison, WI
BS in Computer Science, College of Letters & Science — *Sep. 2026 – May 2030*

George School — Newtown, PA
High School Diploma, College/University Preparatory Program — *Sep. 2021 – May 2026*

- Took gap year 2024-2025 to focus full-time on competitive programming, machine learning, robotics, and heuristic optimization, winning multiple university contests
- **Relevant Coursework:** Advanced Programming: Artificial Intelligence, Statistics with Data Science, Proof & Advanced Topics in Mathematics, Advanced Math: Linear Algebra, AP Calculus BC, Advanced Physical Computing & Robotics

Technical Skills

Languages: C, C++, Java, Python, Swift/SwiftUI, HTML/CSS/JavaScript
Frameworks & Tools: PyTorch, Git, Xcode, iOS Development, CAD, Teensy/Embedded Systems

Projects

Tiaoqi Master – Chinese Checkers iOS App — github.com/markjouh/tiaoqi-master
Swift Student Challenge 2026 Winner — *2025 – 2026*

- Designed and built a native iOS app in Swift 6 and SwiftUI that teaches Chinese Checkers through a play-as-you-learn tutorial and an adaptive AI opponent that scales to the player's skill level
- Implemented an iterative-deepening minimax engine with alpha-beta pruning, heuristic move ordering, and a time-budgeted search over an axial hex-coordinate board for responsive on-device play
- Authored a contextual commentary system that reacts to chain length, forward progress, and setup play, backed by a procedural audio-synthesis engine built on AVFoundation

RotateMate – Image Rotation ML Model — github.com/markjouh/RotateMate-model
Personal Project — *2025 – Present*

- Built an end-to-end iOS ML system to automatically detect and correct mis-rotated images, spanning custom dataset creation, PyTorch training, and Swift/SwiftUI deployment
- Optimized for on-device efficiency: designed 3M parameter model, implemented quantization pipeline, and engineered hardware-accelerated preprocessing to achieve <1ms inference latency on modern NPUs

Competitive Programming Library — github.com/markjouh/cp-library
Open Source Project — *2023 – Present*

- Authored a C++ library of advanced data structure and algorithm implementations for Codeforces, ICPC, and educational reference

Research

Research Assistant, Gerych Lab — Worcester Polytechnic Institute
Trustworthy ML (Remote); Advisor: Prof. Walter Gerych — *Apr. 2026 – Present*

- Contributing to research on test-time debiasing of CLIP-style vision-language models, building on the lab's BendVLM (NeurIPS 2024) and WRING work on concept-subspace bias amplification
- Onboarding via advisor-curated reading on CLIP/VLM internals and concept-subspace debiasing; preparing evaluation work on projection- vs rotation-based methods
- Expected middle-author contribution on a forthcoming paper

Experience

Algorithms Club – President & Co-Founder George School
Leadership Position *Sep. 2024 – Present*

- Co-founded and lead school's Algorithms Club with 30+ active members
- Organize weekly meetings covering competitive programming techniques, algorithm design, and problem-solving strategies
- Helped members advance through USACO divisions, up to Platinum (the contest's highest)

Math Tutor George School
Volunteer Position *Sep. 2024 – Present*

- Provide weekly peer tutoring in advanced mathematics including combinatorics and other topics
- Help students develop problem-solving skills and mathematical intuition through one-on-one instruction

Honors & Awards

Swift Student Challenge Winner Apple
Selected as a 2026 Swift Student Challenge winner for Tiaoqi Master, a SwiftUI app teaching Chinese Checkers *2026*

USACO Platinum USA Computing Olympiad
Highest division of premier US high school algorithmic competition *Mar. 2025*

Codeforces Master (Rating: 2219) Codeforces
Top 1% of competitors worldwide in timed algorithmic contests *Aug. 2024*

First Place – CMIMC Programming Contest Carnegie Mellon University
1st overall and 1st in Optimization round, team-based algorithmic challenges *Apr. 2025*

First Place – Massachusetts Computer Science Olympiad MACSO
Won in-person high school finals with 278 participants from 182 schools in 40 countries *Nov. 2024*

First Place – PClassic Programming Contest (Advanced) University of Pennsylvania
1st overall in Advanced Division, four-hour algorithm and data structure contest *Apr. 2024*

First Place – TJ Invitational Open in Informatics (TJIOI) Thomas Jefferson High School
1st overall in two-person team, multi-hour algorithmic coding challenges *Apr. 2024*